SANJAY P PURUSHOTHAMAN

Forward Deployed Engineer | Applied AI & Agentic Systems | Full-Stack (Next.js, Node.js, AWS)

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PROFESSIONAL SUMMARY

Forward Deployed–oriented engineer with **5+ years of professional experience** across software engineering and digital product delivery, including 4+ years building and shipping production full-stack and cloud systems. Combines a consultant's discipline stakeholder-led problem discovery, requirement framing, solution design and executive delivery with an engineer's end-to-end ownership currently owning the frontend of a multi-service product, joining client requirement calls, and shipping through GitLab CI/CD to AWS (Lambda, EC2, ECS) and Kubernetes. Currently building **agentic AI systems with multi-model LLM orchestration (LangChain)** in the PGP in Forward Deployed Engineering & Applied AI Solutions including a bilingual order-resolution assistant with a tiered model strategy and fail-closed anti-hallucination guardrails, delivered from discovery to MVP.

CORE SKILLS

Applied AI & LLMs: Agentic AI design patterns, multi-model LLM orchestration & routing, LLM guardrails and evaluator/validator patterns, fail-closed design, LangChain, Google Gemini API, OpenAI API, prompt & context engineering, entity extraction, AI workflow automation (n8n, Python), Slack API integrations, RAG (in progress)

Forward Deployed & Consulting: Stakeholder-led problem discovery, requirement discovery, PRDs, solution design documents, MVP scoping, trade-off communication, executive client presentations

Backend: Node.js, Express.js, NestJS, Python API integration, REST, GraphQL, microservices, serverless

Frontend: React.js, Next.js, TypeScript, JavaScript, Vite, Tailwind CSS, Zustand, HTML5, CSS3

Cloud & DevOps: AWS (Lambda, EC2, ECS, CloudWatch, S3, Cognito), Docker, Kubernetes, GitLab CI/CD, Jenkins, GitHub Actions

Data & Security: PostgreSQL, Supabase, MongoDB, OAuth2, JWT, RBAC, input validation

Practices: System design, manifest-driven architecture, design patterns, MVC, Agile/Scrum

FORWARD DEPLOYED ENGINEERING PROJECTS

Dhaga & Co. AI-Powered D2C Order Resolution Assistant | FDE Academy, Cohort 3

- Owned an end-to-end Forward Deployed engagement for a D2C brand, scoping repetitive “Where is my order?” (WISMO) queries — answered in ~9 hours on average — as the core support bottleneck through stakeholder-led discovery.
- Architected a **7-stage agentic LLM pipeline**: intent and sub-intent routing, order-ID extraction with regex fallback, Supabase lookup, structured fact extraction, response generation, validation and fail-closed fallback — replacing the ~9-hour manual turnaround with real-time answers.
- Designed a **tiered two-model strategy**: Gemini Flash-Lite for high-frequency classification, extraction and validation; a higher-tier Gemini Flash model only for bilingual **English/Hinglish** response generation, minimising token cost and latency.
- Built an **anti-hallucination guardrail** (evaluator pattern) that audits every response against database ground truth; on a failed check or timeout, a **fail-closed** engine sends a deterministic templated reply, so unverified claims never reach customers.
- Chose **direct relational lookups over vector RAG** for exact, real-time order data (status, ETA, courier, payment mode), keeping the LLM strictly as the language layer over a deterministic source of truth; defended the architecture and trade-offs to executive stakeholders.
- **Stack:** TypeScript, Node.js, Express, React, Vite, Tailwind CSS, Google Gemini API (@google/genai), Supabase (PostgreSQL)

Slack Team Insights AI Workflow Automation | FDE Academy, Cohort 3

- Built an AI-powered workflow (**Slack → Python → Gemini → Team Insights**) that turns scattered Slack team updates into concise, readable team summaries.
- Integrated **Slack APIs** to collect team updates and wrote **Python automation** to process them and prompt the **Gemini API** for summary generation.
- Brought AI into the tool where the team already works, so insights reach people without changing their workflow.
- Designed it as a **Slack-based AI chatbot workflow**, so team members get summaries and insights directly inside Slack rather than in a separate tool.
- Delivered a working end-to-end prototype within the hackathon timeframe, demonstrating how collaboration platforms can be connected to LLMs to cut the time spent reading through team updates.
- **Stack:** Python, Slack API, Google Gemini API

PROFESSIONAL EXPERIENCE

Associate Consultant | Zimble Systems Pvt. Ltd. *Feb 2026 – Present*

- Own the entire frontend of **Config App**, the company's core product a single Next.js application through which multiple backend services are configured and managed; currently leading its frontend development.
- Participate in **client calls for requirement discussions**, translating client needs into scoped frontend features and delivery priorities.
- Designed and built **HyperConfigurator** from scratch within the Config App ecosystem: a **Node.js serverless backend on AWS Lambda**, a Next.js frontend with Zustand state management, and integration of a specialised Python API.
- Own deployments end to end: **GitLab CI/CD** pipelines, **Docker image** builds, and releases to **AWS EC2, ECS and Kubernetes**.
- Monitor and debug production with **AWS CloudWatch**, using log analysis to diagnose issues across services and sustain uptime.
- Maintain manifest-driven application servers, managing MongoDB integrations and secure GitHub authentication.

Full Stack Developer | JAH Transport LLC *Jun 2025 – Jan 2026*

- Built and deployed an **Inventory & Supply Chain Management** application (Next.js, MongoDB, REST APIs) that digitised daily operational tracking for the business.
- Implemented **role-based access control (RBAC)** across the admin dashboard, securing data and tailoring functionality to each department.

Junior Software Engineer | Softnotions Technologies Pvt. Ltd. *Mar 2023 – Jun 2025*

- Shipped **AI-powered features**, including contextual content suggestions using the **OpenAI API**, integrated directly into React interfaces.
- Developed full-stack applications with Next.js, React.js and Node.js, improving frontend responsiveness and backend efficiency.
- Designed REST and GraphQL APIs powering dynamic UI components and third-party integrations.
- Containerised Node.js services with **Docker** and built **Jenkins** CI/CD pipelines for consistent staging and production releases.
- Built real-time analytics dashboards and enforced security with OAuth2, JWT and strict input validation.

MERN Stack Developer Trainee | Prepbytes (Remote) *Jul 2022 – Mar 2023*

- Built CRUD blog platforms and secure admin dashboards (MongoDB, Express.js, React) with JWT authentication, validation and multi-format file uploads, working in Agile sprints with peer code reviews.

Graphic Designer | Calypso Branding Partner *Jul 2021 – Jun 2022*

- Designed high-fidelity UI layouts and built responsive, semantic marketing landing pages (HTML5, CSS3), developing the user-centred design sense used in client-facing solution work today.

EDUCATION

PGP in Forward Deployed Engineering & Applied AI Solutions | FDE Academy, Bengaluru Aug *2026 – Present*

Selective program (≈4% acceptance). Dual-track: production AI engineering (LLM apps, agents, RAG, evaluation, deployment) and consulting (discovery, PRDs, solution design, client delivery). Hands-on labs: LangChain use cases (LLM workflows, chains and agents); currently studying advanced RAG.

MBA in Artificial Intelligence | Amrita Vishwa Vidyapeetham April *2026 – Present*

B.Tech in Electronics and Communication Engineering | APJ Abdul Kalam Technological University *2017 – 2021*

CERTIFICATIONS

- **Certificate — Dhaga & Co. Client Engagement** (Agentic Design Patterns, LLM Orchestration, Problem Discovery, MVP Engineering), FDE Academy, Oct 2026
- **Generative AI Certification** — Skill India & National Skill Development Corporation (NSDC)

LANGUAGES

English, Malayalam, Hindi, Tamil